

SPELLZEE ERP

Data Architecture & Data Dictionary

Draft 1

Business-Level Entity, Field, Relationship, Lifecycle & Governance Specification

Purpose: Backend/database foundation derived from product discovery

Status: Working Draft — to be validated before technical schema design

This document describes what information the Spellzee ERP needs to know and how major business records relate. It intentionally avoids prescribing a specific database technology, SQL schema or implementation framework.

1. How to Use This Document

Draft 3 is the product/business overview. This document is the corresponding data foundation. Product requirements answer what the ERP must do; this document answers what the ERP must remember, identify, connect, change, protect and report.

- Business entity = a meaningful real-world object such as Student, Teacher, Enrollment or Session.
- Record = one instance of an entity.
- Field = a property stored about a record.
- Relationship = how two entities connect.
- Event/history = something that happened without destroying previous state.
- Status = current lifecycle state.
- Identifier = stable key used to reference a record.

Important: The final physical database design should be produced by engineering after validating this business model. Fields marked as examples are not automatically mandatory.

2. Global Data Architecture Principles

- Every major entity receives a Spellzee-generated stable ID.
- Do not use names, phone numbers or email addresses as primary identifiers.
- External IDs are stored separately and mapped to Spellzee IDs.
- Student identity is permanent; enrollments, subscriptions, payments, schedules and sessions are separate records.
- Historical changes are preserved rather than silently overwritten.
- Critical records are normally cancelled/archived rather than deleted.
- Sensitive changes are permission-controlled and auditable.
- Dates should distinguish created, updated, effective-from and effective-to where applicable.
- Records should carry source/context when created by an integration or workflow.
- Relationships should be designed so a complete Student 360 can be reconstructed.
- AI-generated fields are derived information, not authoritative source records.

3. Core Entity Map

High-level relationship:

Parent/Guardian → Student → Enrollment → Subscription → Payment

Enrollment → Class Schedule → Sessions → Attendance / Lesson / Recording

Student → Coordinator Ownership / Tickets / Communications / Retention

Teacher → Availability / Allocation / Sessions / Training / Observation / Incentive / Payroll

Teacher + Capacity → Demand Forecast → Recruitment Requirement

4. Identifier Standards

Entity	Suggested ID	Purpose
Student	STU-YYYY-NNNNNN	Permanent student identity.
Parent/Guardian	PAR-YYYY-NNNNNN	Permanent parent/guardian identity.
Employee	EMP-YYYY-NNNNNN	Permanent employee identity.
Teacher	TCH-YYYY-NNNNNN	Permanent teacher identity; may map to Employee.
Enrollment	ENR-YYYY-NNNNNN	Specific academic participation.
Subscription	SUB-YYYY-NNNNNN	Commercial session/service entitlement.
Payment	PAY-YYYY-NNNNNN	Individual financial transaction.
Class Schedule	CLS-YYYY-NNNNNN	Recurring/planned class arrangement.

Entity	Suggested ID	Purpose
Session	SES-YYYY-NNNNNN	Individual class occurrence.
Ticket	TKT-YYYY-NNNNNN	Operational issue/request.
Observation	OBS-YYYY-NNNNNN	Teacher quality observation.
Training	TRN-YYYY-NNNNNN	Training participation/program record.
Approval	APR-YYYY-NNNNNN	Controlled approval request.
Communication	COM-YYYY-NNNNNN	Conversation/message record where required.
Call	CAL-YYYY-NNNNNN	Call activity record.
Audit Event	AUD-YYYY-NNNNNN	Immutable/controlled audit event.

5. Parent / Guardian Entity

5.1 Purpose

Represents the adult responsible for one or more students and the customer communication relationship.

5.2 Core Fields

Field	Type / Concept	Notes
Parent ID	Permanent identifier	Never reused.
Full name	Text	Primary display name.
Primary phone	Contact	May change; history retained.
Alternate phone	Contact	Optional.
Email	Contact	May change; historical values retained.
Address	Structured text	Address history may be needed.
Relationship to student	Reference/value	Parent, guardian, etc.
Preferred language	Value	Useful for communication.
Communication preference	Value	WhatsApp, in-app, email, etc.
Status	Lifecycle	Active/inactive.
Created/updated	Timestamp	System audit.
Notes	Controlled text	Operational notes subject to permissions.

One parent can be linked to multiple students. A student can have multiple authorized parents/guardians. The relationship itself should be a separate record if permissions, primary contact or validity dates need to be stored.

6. Student Entity

6.1 Core Identity Fields

Field	Type / Concept	Notes
Student ID	Permanent identifier	Primary Spellzee identity.
Full name	Text	Do not use as unique key.
Preferred/display name	Text	Optional.
Date of birth	Date	Use according to policy.
Gender	Controlled value	Only if operationally required.
Primary parent/guardian	Relationship	Primary contact.
Other guardians	Relationship	Multiple permitted.
Phone/contact references	Relationship	Usually parent contact; avoid duplicating unnecessarily.
Email/contact references	Relationship	Historical values may remain.

Field	Type / Concept	Notes
Address	Relationship/value	Usually parent/household address.
Preferred language	Value	Relevant to service.
Learning language	Value	Relevant to matching.
Student status	Lifecycle	Lead/Prospect is preferably handled separately; examples include Active, Break, Completed, Cancelled, Inactive.
Source	Value/reference	Origin such as admission system/import.
Created at	Timestamp	System generated.
Updated at	Timestamp	System generated.

6.2 Operational Fields / Links

- Current enrollment(s).
- Current subscription(s).
- Current teacher(s).
- Current class schedule(s).
- Current coordinator/owner.
- Student health/risk state.
- Attendance summary.
- Academic progress summary.
- Open tickets.
- Retention status.
- Previous names/contact identifiers where permitted.
- External-system IDs.

6.3 Student Identity Rules

- A student ID must not change when contact details change.
- Returning students create a new enrollment/subscription, not a new student identity.
- Potential duplicates must be surfaced before creation.
- Merges must preserve source records and redirect relationships safely.

7. Student Contact & Identity History

Because phone/email/name can change, these should not be treated as the permanent identity.

Record	Fields	Purpose
Contact History	Student/Parent, type, value, effective-from, effective-to, primary flag, verified flag	Preserve historical contact details.
Identity Match	Candidate records, match signals, confidence, decision, reviewer	Support duplicate prevention.
Merge Event	Source student, target student, reason, approver, date	Preserve duplicate resolution history.

8. Enrollment Entity

Enrollment represents participation in a particular course/program and is separate from the student and commercial subscription.

- Enrollment ID.
- Student ID.
- Program/course.

- Subject.
- Level.
- Learning mode/session type.
- Start date.
- Expected end date.
- Actual end date.
- Status.
- Assigned academic context.
- Current class schedule.
- Teacher assignment.
- Reason for cancellation/pause/completion.
- Created/updated timestamps.

A student may have multiple historical and active enrollments.

9. Subscription Entity

Subscription represents the commercial entitlement purchased by the student/parent.

- Subscription ID.
- Student ID.
- Enrollment ID where applicable.
- Plan/product.
- Purchased session count.
- Price/currency.
- Purchase date.
- Start date.
- Validity/expiry.
- Sessions scheduled.
- Sessions completed.
- Sessions consumed.
- Sessions protected.
- Sessions remaining.
- Credits/adjustments.
- Status.
- Renewal information.
- Payment references.
- Cancellation/refund references.

The subscription should not be used as the student identity and should not replace individual payment records.

10. Payment Entity

- Payment ID.
- Student ID.
- Parent/customer reference.
- Subscription ID.
- Enrollment reference where applicable.

- Amount.
- Currency.
- Payment method.
- Gateway/reference ID.
- Payment date/time.
- Status.
- Invoice/receipt reference.
- Refund/adjustment references.
- Created by/source.
- Approval status where relevant.
- Audit history.

Financial records should be append-oriented where possible: corrections should create controlled adjustments rather than silently rewriting settled transactions.

11. Teacher Entity

Field	Type / Concept	Purpose
Teacher ID	Permanent identifier	Stable teacher identity.
Employee ID	Reference	Connect to HR if applicable.
Full name	Text	Display.
Contact details	Contact	Phone/email; history as needed.
Qualifications	Structured data	Credentials.
Experience	Structured/text	Professional background.
Subjects	Many-to-many	Teaching capability.
Levels	Many-to-many	Supported student levels.
Languages	Many-to-many	Language capability.
Bilingual capability	Capability	Tamil-English, Hindi-English, etc.
Specializations	Many-to-many	Skills used for matching.
Employment status	Lifecycle	Onboarding, active, inactive, exited.
Training status	Reference	Training/certification state.
Capacity settings	Reference	Availability/capacity rules.
Payroll reference	Restricted reference	Finance/HR.
Documents	Document references	Controlled access.

12. Teacher Availability & Capacity

Availability and capacity are related but distinct.

12.1 Availability Record

- Teacher ID.
- Date or recurring day.
- Start/end time.
- Availability type: regular, specific-date, temporary, unavailable.
- Effective-from/to.
- Reason.

- Source.
- Approval if required.

12.2 Capacity Record

- Teacher/skill category.
- Time slot.
- Maximum planned capacity.
- Allocated capacity.
- Free capacity.
- Temporary released capacity.
- Reserved capacity.
- Utilization.
- Effective period.

12.3 Forecast Capacity

- Current free slots.
- Expected releases from breaks.
- Expected releases from subscription completion.
- Expected teacher return.
- Forecast demand.
- Projected capacity gap/surplus.
- Confidence/assumption metadata.
- Forecast period.

13. Teacher Allocation Entity

- Allocation ID.
- Student ID.
- Enrollment ID.
- Teacher ID.
- Class Schedule ID.
- Allocation type.
- Start/effective date.
- End date.
- Previous teacher/schedule references where relevant.
- Reason.
- Requested by.
- Approved by.
- Status.
- Created/updated timestamps.

Allocation is not merely a teacher assignment. It represents the teacher + schedule arrangement for delivery.

14. Coordinator Ownership Entity

- Ownership ID.
- Student ID.

- Parent ID where relevant.
- Employee/coordinator ID.
- Ownership role.
- Start date/time.
- End date/time.
- Reason for transfer.
- Current flag.
- Escalation level.
- Related workflow/ticket.

This supports onboarding ownership, first-month coordination, Student Success, Retention, Operations, Academic and Ticket ownership without overwriting history.

15. Class Schedule Entity

- Class Schedule ID.
- Enrollment ID.
- Student/group reference.
- Teacher ID(s).
- Session type: 1-to-1/group.
- Subject/course.
- Days of week.
- Start/end time.
- Timezone.
- Start date.
- Planned end date.
- Recurring/permanent indicator.
- Status.
- External Merithub class ID.
- Host/common links as appropriate.
- Schedule version/history.

Recurring schedule is distinct from individual sessions.

16. Session Entity

- Session ID.
- Class Schedule ID.
- Student/enrollment.
- Teacher.
- Scheduled start/end.
- Actual start/end.
- Session type.
- Session purpose: regular, compensation, replacement, extra, etc.
- Status/outcome.
- Cancellation reason.
- Reschedule reference.

- Compensation required flag.
- Credit impact.
- External Merithub session/class reference.
- Recording reference.
- Created/updated timestamps.

16.1 Session Outcome Examples

- Completed/attended.
- Partial/late/absent according to policy.
- Parent advance cancellation.
- Late cancellation.
- Teacher absent.
- Teacher-side technical issue.
- Student-side technical issue.
- Spellzee cancellation.
- No-show.
- Rescheduled.
- Compensation required/completed.

17. Attendance Entity

- Attendance ID.
- Session ID.
- Student ID.
- Teacher ID.
- Scheduled minutes.
- Actual attended minutes.
- Attendance percentage.
- Attendance status.
- Join/leave timestamps where available.
- Source system.
- Manual correction flag.
- Correction reason.
- Corrected by/approved by.
- Created timestamp.

Example: a 60-minute session with 55 attended minutes can calculate 91.7% attendance. Thresholds for Present/Late/Partial/Absent remain policy decisions.

18. Compensation / Makeup Entity

- Compensation ID.
- Original session ID.
- Student/enrollment.
- Reason.
- Eligibility decision.
- Credit protected/consumed.

- Requested by.
- Approved by if required.
- Teacher.
- Compensation session ID.
- Scheduled date/time.
- Status.
- Validity/expiry.
- Number of reschedules.
- Outcome.
- History.

Compensation must not alter the original recurring schedule. It is an additional linked session.

19. Reschedule / Change Request Entity

- Request ID.
- Student/enrollment/session.
- Requested change type.
- Old schedule/session.
- Requested new date/time.
- Reason.
- Requested by.
- Teacher availability result.
- Coordinator decision.
- Approval.
- Status.
- Effective date.
- Communication references.
- Audit history.

This provides a controlled trail for parent requests, teacher unavailability and operational changes.

20. Academic Program / Curriculum

- Program ID.
- Course.
- Subject.
- Level.
- Curriculum version.
- Units/modules.
- Learning objectives.
- Sequence.
- Active dates.
- Status.
- Owner/version metadata.

Exact curriculum structure remains a discovery item.

21. Lesson Entity

- Lesson ID.
- Session ID.
- Enrollment.
- Teacher.
- Date/time.
- Subject/topic.
- Curriculum unit.
- What was taught.
- Activities.
- Homework/outcome.
- Teacher notes.
- Student response/progress notes.
- Materials references.
- Status.

22. Assessment Entity

Assessment should be typed by purpose rather than becoming one generic record.

Assessment Type	Purpose	Primary Module
Demo Assessment	Establish starting level before/around admission	Admissions / Academic
Student Academic Assessment	Measure learning performance	Academic
Teacher Observation	Measure teaching quality	Teacher Performance
Training Assessment	Measure employee/teacher training readiness	HR / L&D;
Sales/Pitch Assessment	Evaluate employee pitch/call performance	HR / L&D; / Sales

Assessment fields can include assessment ID, subject/person, assessor, date, criteria, score, feedback, evidence/reference, outcome, action plan and review date.

23. Student Progress Entity

- Progress ID.
- Student/enrollment.
- Subject/skill.
- Baseline.
- Current level.
- Progress measure.
- Evidence/assessment references.
- Teacher/academic reviewer.
- Strengths.
- Areas for improvement.
- Next goals.
- Last reviewed.
- Status/history.

24. Learning Materials / Worksheets / Activities

- Material ID.
- Title.
- Type: worksheet, document, video, activity, link, etc.
- Subject/topic.
- Level.
- Curriculum mapping.
- Author/owner.
- Version.
- Access permissions.
- Student/enrollment assignment.
- Assigned date.
- Completion/submission where applicable.
- Status.
- File/link reference.

25. Recording Entity

- Recording ID.
- Session ID.
- Merithub recording ID/link.
- Start/end.
- Availability status.
- Access permission.
- Retention/expiry.
- Processing status.
- Created timestamp.

Recording access must be governed by privacy and product policy.

26. Teacher Training & Certification Data

- Training program ID.
- Teacher/employee ID.
- Program/module.
- Trainer.
- Start/end dates.
- Attendance/progress.
- Assessment results.
- Mock class reference.
- Pass/fail.
- Remediation.
- Certification ID.
- Certification date.
- Expiry/review if applicable.

- Eligibility for allocation.

27. Teacher Observation & Quality Data

- Observation ID.
- Teacher ID.
- Session ID.
- Observer/trainer.
- Round.
- Observation date.
- Criteria set/version.
- Criterion scores.
- Overall score.
- Comments.
- Strengths.
- Improvement areas.
- Action plan.
- Re-observation date.
- Outcome.
- Incentive eligibility reference.

Criteria versions should be preserved so an old observation remains interpretable even if the evaluation rubric changes later.

28. Incentive Entity

- Incentive ID.
- Teacher/employee.
- Period.
- Rule/version.
- Eligibility inputs.
- Performance score.
- Eligible percentage/amount.
- Approval.
- Finance status.
- Payment/payroll reference.
- Adjustment/reversal history.

29. HR Employee Entity

- Employee ID.
- Name/contact.
- Department.
- Role/title.
- Joining date.
- Employment status.
- Manager.
- Documents.

- Training records.
- Leave records.
- Payroll reference.
- Access/role assignments.
- Exit date/reason where applicable.

Teacher can be a specialized employee role, but the data model should support teachers who are not otherwise treated exactly like office employees if Spellzee's employment model requires it.

30. Leave / Unavailability Entity

- Leave ID.
- Employee/teacher.
- Type.
- Start/end.
- Reason/category.
- Requested by.
- Approved by.
- Status.
- Capacity impact.
- Affected allocations/sessions.
- Replacement workflow reference.

31. Ticket Entity

Field	Purpose
Ticket ID	Permanent issue/request identifier.
Student/Parent	Affected customer.
Category	Class, teacher, academic, schedule, subscription, general, etc.
Priority	Business urgency.
Description	Issue/request.
Assigned team	Responsible group.
Assigned person	Responsible owner.
SLA	Target and timestamps.
Status	Open, in progress, resolved, escalated, etc.
Communication	Linked conversation/messages.
Resolution	Outcome.
Related entities	Session, teacher, payment, subscription, etc.
History	All important changes.

32. SLA Entity

- SLA ID.
- Policy/rule.
- Trigger event.
- Ticket/workflow reference.
- Owner.

- Start timestamp.
- Target timestamp.
- Warning timestamp.
- Escalation levels.
- Pause conditions if allowed.
- Breach timestamp.
- Resolution timestamp.
- Status.

The SLA engine should support different policies rather than hard-coding one universal duration. The currently discussed ticket target is 48 hours, pending final approval.

33. Communication / Conversation Data

- Conversation ID.
- Parent/student.
- Participants.
- Channel: WhatsApp, in-app, email, call-linked, etc.
- Owner.
- Created/updated.
- Messages/events.
- Related ticket.
- Related session/enrollment/payment where relevant.
- Status.
- Transfer/ownership history.

A communication may exist without a ticket; a ticket can reference one or more communication records.

34. Call Activity Entity

- Call ID.
- Student/parent.
- Coordinator/employee.
- Provider/external call ID.
- Purpose: onboarding, retention, support, etc.
- Start time.
- Duration.
- Outcome.
- Attempt number.
- Notes.
- Related ticket/task.
- Recording reference if permitted.

FreeJump integration should populate what the provider/API reliably supplies.

35. Retention Interaction Entity

- Retention interaction ID.
- Student/subscription.

- Trigger: session count, expiry proximity, risk, etc.
- Owner.
- Date.
- Attendance/context snapshot.
- Feedback received.
- Concern.
- Action.
- Ticket created.
- Next follow-up.
- Outcome.
- Health/risk change.

36. Student Health / Risk Entity

- Health record ID.
- Student/enrollment/subscription.
- Health state: Healthy, Watch, At Risk.
- Signals.
- Attendance indicators.
- Rescheduling frequency.
- Complaints/tickets.
- Teacher changes.
- Progress indicators.
- Break status.
- Subscription proximity.
- Recommended action.
- Owner.
- Review date.
- History.

AI may later recommend risk state, but the authoritative record should retain the evidence and human decision where required.

37. Capacity & Demand Data Model

- Teacher capacity profile.
- Availability slots.
- Allocated slots.
- Reserved slots.
- Temporary released slots.
- Subject/level/language capability.
- Time-slot capacity.
- Demand forecast.
- Expected admissions.
- Capacity gap.
- Recruitment requirement.
- Forecast period.

- Assumptions/confidence.
- Actual versus forecast outcome.

Capacity should support dimensions such as English, Tamil-English bilingual, Hindi-English bilingual, level, subject, 1-to-1/group and time window.

38. Notification Entity

- Notification ID.
- Recipient.
- Student/parent/teacher/employee.
- Channel.
- Template.
- Trigger.
- Scheduled time.
- Sent time.
- Delivery status.
- Read/response status where available.
- Related entity.
- Retry/error status.

Examples: class reminder, allocation confirmation, compensation confirmation, ticket assignment, SLA warning, retention task and capacity alert.

39. Approval Request Entity

- Approval ID.
- Requested action.
- Entity/type.
- Requester.
- Approver role/person.
- Requested value/change.
- Old value where applicable.
- Reason.
- Supporting evidence/document.
- Status.
- Requested/approved/rejected timestamps.
- Decision reason.
- Audit reference.

Requester and approver should be different for actions covered by maker-checker policy.

40. Roles & Permissions Data

- Role ID/name.
- Permission/action.
- Entity/module.
- View/create/edit/delete/archive/approve/export rights.
- Scope: own/team/department/all.

- Approval authority.
- Effective dates.
- User assignment.
- Exceptions/overrides.

Permission should be separated from business hierarchy. Seniority alone should not grant financial or sensitive-data access.

41. Audit Event Entity

- Audit ID.
- Timestamp.
- User/service.
- Action.
- Entity/type.
- Record ID.
- Old value.
- New value.
- Reason.
- Source/workflow.
- Approval reference.
- IP/device/session metadata where appropriate.
- Outcome: success/blocked/failed.
- Correlation/integration ID.

Audit records for critical actions should be protected from ordinary editing/deletion.

42. External ID Mapping Entity

Field	Purpose
Mapping ID	Internal mapping record.
Spellzee entity/type	Core record.
Spellzee ID	Core identity.
External system	Merithub, TeliCRM, Delicio, FreeJump, payment provider, etc.
External ID	Provider identifier.
Status	Active/revoked.
Created/updated	Mapping lifecycle.
Last sync	Integration monitoring.
Sync status/error	Operational visibility.

43. Key Business Rules Across the Data Model

- One permanent Student ID per person.
- One Payment ID per financial transaction.
- A student can have multiple enrollments and subscriptions over time.
- A subscription has a defined entitlement; individual sessions consume or protect that entitlement according to policy.
- A recurring schedule is not the same as a session.
- A compensation session is linked to an original affected session.

- Teacher changes create history, not overwrite-only updates.
- Coordinator ownership changes create history.
- Teacher availability is not the same as capacity.
- Current capacity plus forecast releases can be compared with demand.
- Critical financial/history records should not be silently deleted.
- Duplicate merges must preserve references and audit trail.
- A teacher should not be allocated if mandatory onboarding/certification rules are not satisfied.
- Approval rules must prevent self-approval where maker-checker applies.
- External IDs never replace Spellzee core IDs.

44. Suggested Data Relationship Scenarios

Scenario A — New Student

Parent → Student → Enrollment → Subscription → Payment → Class Schedule → Sessions → Attendance/Lessons → Progress.
Coordinator ownership and communication attach to the student/workflow.

Scenario B — Returning Student

Existing Student ID is found → new enrollment/subscription may be created → new schedule/teacher if needed → previous academic/operational history remains available.

Scenario C — Teacher Change

Existing allocation ends → new allocation begins → schedule/session relationships update → historical teacher remains in allocation history → next teacher receives structured student history.

Scenario D — Parent Misses Class

Session outcome recorded → policy determines credit impact → if eligible, compensation record created → compensation session scheduled separately → original schedule remains unchanged.

Scenario E — Teacher Absence

Session outcome = teacher absence → entitlement protected according to policy → compensation required → replacement session scheduled → teacher/parent notifications → audit.

Scenario F — Student Break

Enrollment/subscription enters break-related state according to policy → affected schedules/sessions are handled → teacher capacity reflects temporary release → resume later without creating a duplicate student.

Scenario G — Duplicate Attempt

New student request → identity matching → possible existing student shown → user selects existing or requests controlled creation → if merged, all historical relationships are retained.

Scenario H — Capacity Gap

Teacher availability/capacity + breaks + expected releases → compare against demand → identify skill/language/time-slot gap → management/HR receives recruitment or reallocation signal.

45. Data Ownership & Source of Truth

Data	Preferred Owner
Student core identity	Spellzee
Parent identity	Spellzee
Enrollment	Spellzee
Subscription/operational entitlement	Spellzee, with Finance integration
Payment settlement	Finance/payment provider mapped into Spellzee

Data	Preferred Owner
Classroom infrastructure	Merithub, mapped to Spellzee
Teacher profile/HR	Spellzee HR
Teacher quality	Spellzee
Academic record	Spellzee Academic
Communication history	Spellzee communication layer with channel integrations
Calling metadata	Spellzee mapped from provider
External identifiers	Spellzee mapping layer

46. Data Lifecycle & Retention

- Define which entities can be archived.
- Define retention period for financial records, recordings, communications and audit logs.
- Define what happens to data after student cancellation/completion.
- Define privacy rules for parent/child data.
- Define who can view historical records after role changes.
- Define backup retention separately from application-level archival.
- Never design deletion without considering dependent records and audit requirements.

47. Data Quality & Validation Framework

- Required-field validation.
- Unique identifier validation.
- Reference integrity.
- Date consistency.
- Status-transition validation.
- Duplicate detection.
- Contact verification where appropriate.
- Financial amount/currency validation.
- Session entitlement reconciliation.
- External ID uniqueness per provider.
- Automated anomaly detection later.

48. Reporting Views the Data Model Must Support

- Student 360.
- Student history.
- Teacher 360.
- Coordinator workload.
- Current teacher capacity.
- Forecast capacity.
- Capacity by language/subject/level/time.
- Subscription consumption.
- Planned versus projected completion.
- Compensation backlog.

- Attendance trends.
- Teacher observation/performance.
- Training/certification status.
- Ticket/SLA performance.
- Retention health.
- Payment/renewal forecast.
- Audit/change history.

49. AI-Derived Data — Keep Separate From Source Data

AI summaries, risk predictions, teacher/student matching scores and forecasts should be stored as derived outputs with model/version/timestamp/source references where practical. They should not overwrite the source lesson, assessment, attendance, payment or identity records.

- AI summary ID.
- Entity/reference.
- Generated timestamp.
- Model/version.
- Source data snapshot/reference.
- Output.
- Confidence/quality indicator if applicable.
- Human review/status.
- Expiry/recompute policy.

50. Recommended Backend Design Sequence

- Validate this business data dictionary with management and domain owners.
- Convert entities into a formal logical data model.
- Define cardinality: one-to-one, one-to-many, many-to-many.
- Define mandatory/optional fields.
- Define lifecycle/status transitions.
- Define reference/master-data tables.
- Define permission and audit requirements.
- Define integration mappings.
- Define physical database schema and indexes.
- Define API contracts.
- Define migration strategy from current tools.
- Define test data and data-quality checks.

Do not start by creating hundreds of tables from this document. First validate the business entities and relationships. Some entities may later be implemented as tables, event records, views, JSON structures or other technical mechanisms depending on engineering decisions.

51. Open Data Decisions

- Exact required fields for student/parent privacy and operational need.
- Whether Student and Employee records are separate or share a person master.
- Exact teacher employment/contract model.
- Exact session credit rules and reconciliation logic.

- Exact cancellation/compensation policy.
- Exact academic curriculum model.
- Exact assessment rubric structures.
- Document/file storage strategy.
- Recording retention and access.
- Financial accounting integration.
- External system migration and historical-data strategy.
- Data privacy/consent requirements.
- Audit retention period.
- Which fields may be corrected without approval.
- Capacity unit: minutes, hours, sessions or weighted slots.
- Forecast data granularity and confidence model.

52. Final Data Architecture Principle

The database should represent the business truth, not the current spreadsheet structure.

If the business changes teacher, schedule, coordinator, subscription or course, the data model should preserve the relationship history. If a parent returns with a different phone number, the system should still recognize the student. If a session is missed, the system should know why, whether credit was consumed, whether compensation is owed and how the projected completion date changed. If capacity changes because students go on break, management should be able to see the released capacity and future demand.

This data foundation is intended to support the Spellzee ERP described in the Master Product & Business Requirements Draft 3 and should be validated before engineering freezes the physical database design.